Discrete Math W6

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Hello

I am Diego

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Group Chat



Plan

1. Feedback
2. Theory Recap
3. Exercises
4. Questions?

Feedback

Theory Recap

Where Are We?

- Basic Logic & Proofs
- ⇒ Sets & Relations
- Number Theory
- Algebra
- Logic

3. Sets & Relations

- Introduction
- Sets and Operations on Sets
- Relations
- Equivalence Relations
- Partial Order Relations
- Functions
- Countable and Uncountable Sets

Questions?

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Functions

Definition 3.33

A *function* $f : A \to B$ from a *domain* A to a *codomain* B is a relation from A to B with the special properties (using the relation notation $a\ f\ b$):

1. $\forall a \in A\ \exists b \in B \quad a\ f\ b$ (f is totally defined),
2. $\forall a \in A\ \forall b, b' \in B\ \left(a\ f\ b \wedge a\ f\ b' \to b = b'\right)$ (f is well-defined).

Definition 3.34

The set of all functions $A \to B$ is denoted as B^A.

$\mathbb{N}^{\mathbb{N}}$ are all functions from $\mathbb{N}$ to $\mathbb{N}$

Partial functions

Definition 3.35

A *partial function* $A \to B$ is a relation from A to B such that condition 2. above holds.

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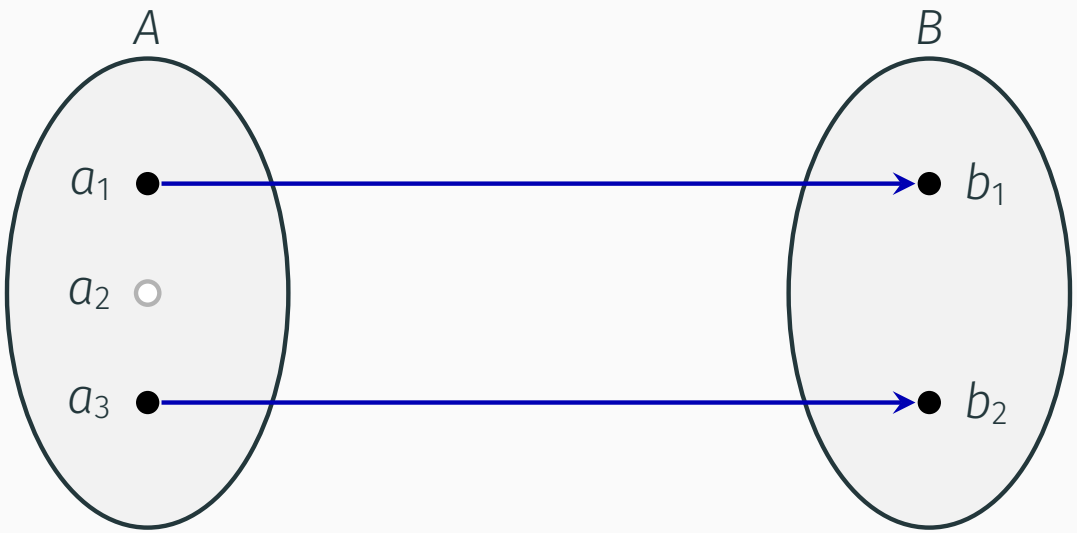


Image / Range / Preimage

Definition 3.36

For a function $f : A \to B$ and a subset S of A, the *image* of S under f, denoted $f(S)$, is the set

$$f(S) \stackrel{\text{def}}{=} \{f(a) \mid a \in S\}.$$

Definition 3.37

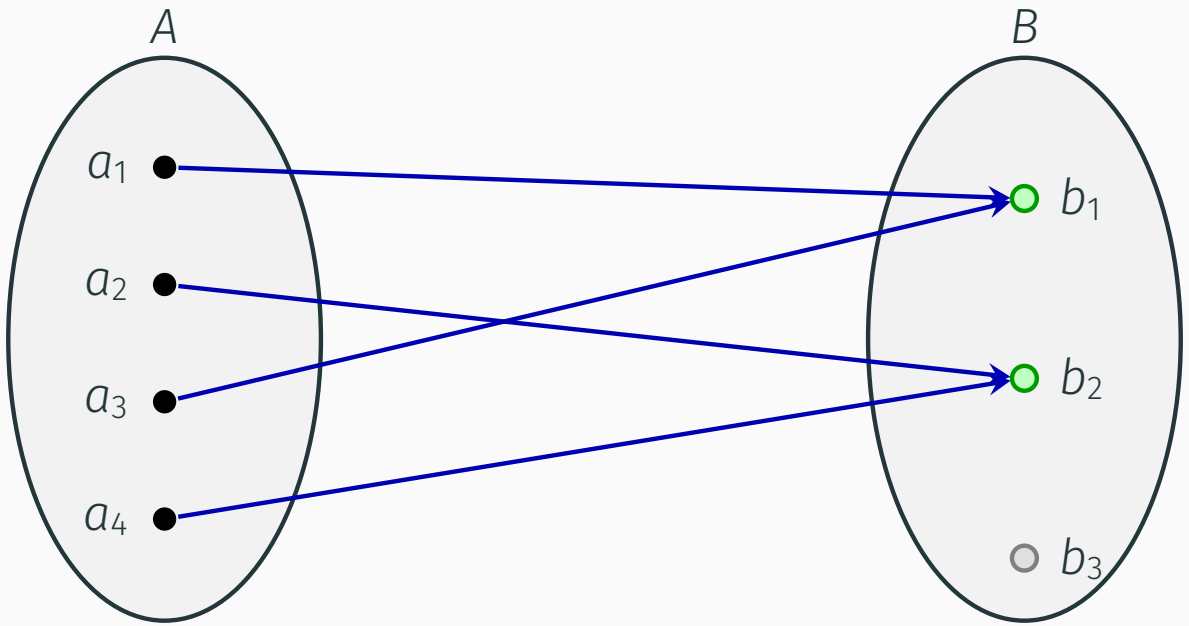
The subset $f(A)$ of B is called the *image* (or *range*) of f and is also denoted $\mathsf{Im}(f)$.

Definition 3.38

For a subset T of B, the *preimage* of T, denoted $f^{-1}(T)$, is the set of values in A that map into T:

$$f^{-1}(T) \stackrel{\text{def}}{=} \{a \in A \mid f(a) \in T\}.$$

Image, Range and Preimage



Range / Image of A: $\mathrm{Im}(f) = f(A) = \{b_1, b_2\} \subsetneq B$ (green)

Image, Range and Preimage

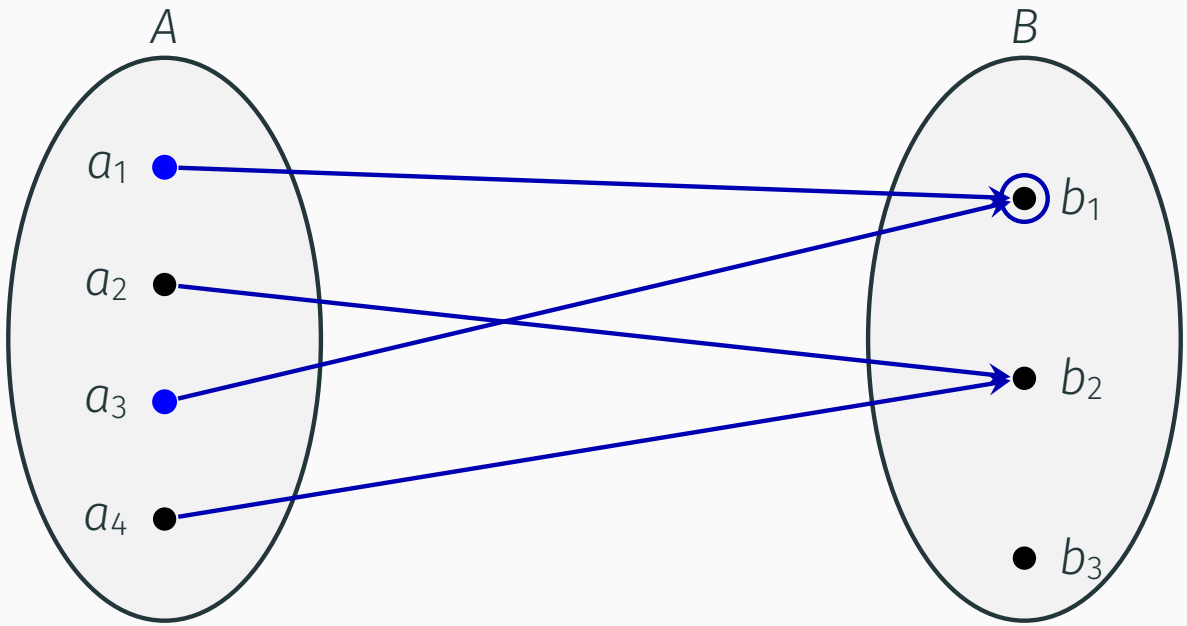
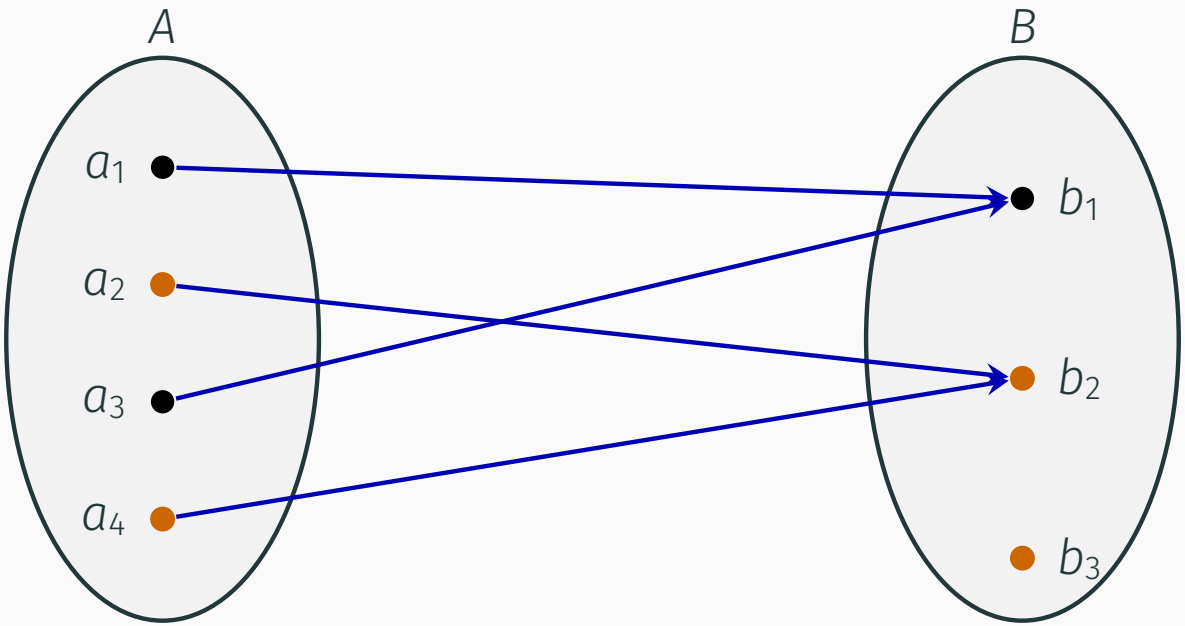


Image of $S = \{a_1, a_3\}$: $f(S) = \{b_1\}$ (blue)

Image, Range and Preimage



Preimage of $T = \{b_2, b_3\}$: $f^{-1}(T) = \{a_2, a_4\}$ (orange)

Injective etc

Definition 3.39

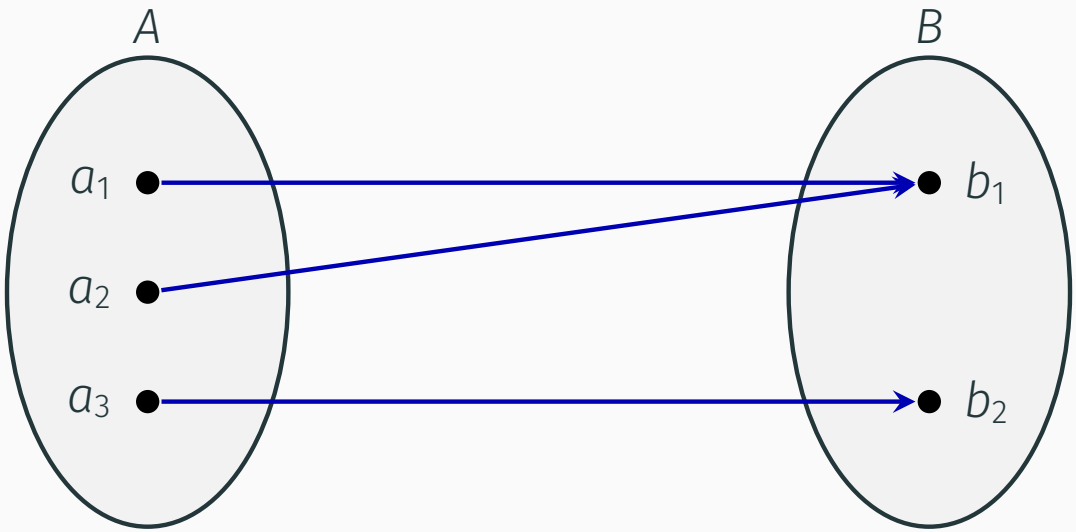
A function $f : A \to B$ is called

1. *injective* (or *one-to-one* or an *injection*) if for $a \neq a'$ we have $f(a) \neq f(a')$, i.e., no two distinct values are mapped to the same function value (there are no "collisions").
2. *surjective* (or *onto*) if $f(A) = B$, i.e., if for every $b \in B$, $b = f(a)$ for some $a \in A$ (every value in the codomain is taken on for some argument).
3. *bijective* (or a *bijection*) if it is both injective and surjective.

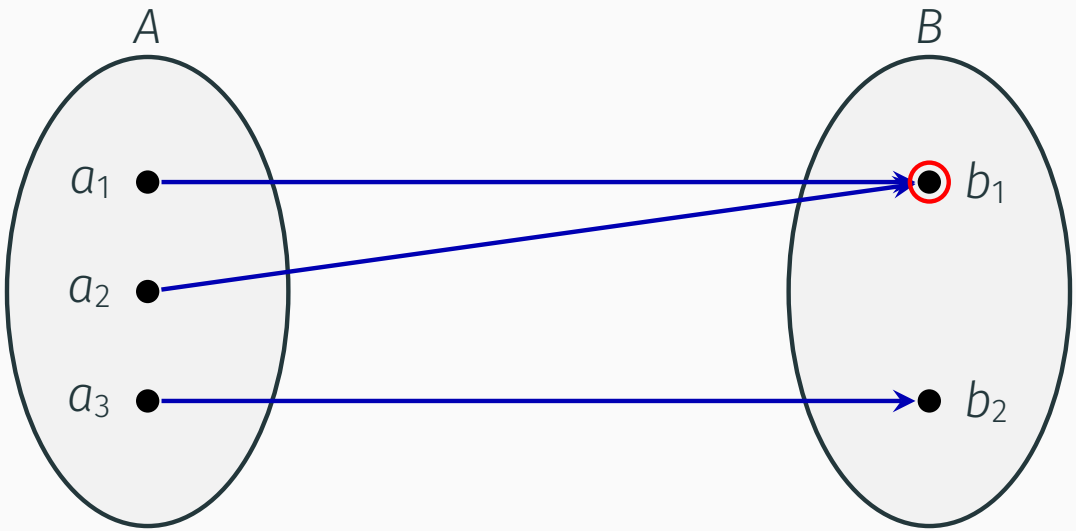
Definition 3.40

For a bijective function f, the *inverse* (as a relation, see Definition 3.11) is called the *inverse function* of f, usually denoted as f^{-1}.

Example



Example



$a_1 \neq a_2$ but $f(a_1) = f(a_2) = b_1$ not injective

function composition

Definition 3.41

The *composition* of a function $f : A \to B$ and a function $g : B \to C$, denoted by $g \circ f$ or simply gf, is defined by $(g \circ f)(a) = g(f(a))$.

Lemma 3.14

Function composition is associative, i.e., $(h \circ g) \circ f = h \circ (g \circ f)$.

Countability of sets

Definition 3.42

(i) Two sets A and B *equinumerous*, denoted $A \sim B$, if there exists a bijection $A \to B$.

(ii) The set B *dominates* the set A, denoted $A \preceq B$, if $A \sim C$ for some subset $C \subseteq B$ or, equivalently, if there exists an injective function $A \to B$.

(iii) A set A is called *countable* if $A \preceq \mathbb{N}$, and *uncountable* otherwise.

Between finite and countably infinite

Lemma 3.15

(i) The relation $\sim$ is an equivalence relation.

(ii) The relation $\preceq$ is transitive: $A \preceq B \wedge B \preceq C \implies A \preceq C$.

(iii) $A \subseteq B \implies A \preceq B$.

Theorem 3.16

$A \preceq B \wedge B \preceq A \implies A \sim B$.

Theorem 3.17

A set A is countable if and only if it is finite or if $A \sim \mathbb{N}$.

Important countable sets

Theorem 3.18

The set $\{0,1\}^* \stackrel{\text{def}}{=} \{\epsilon, 0, 1, 00, 01, 10, 11, 000, 001, \ldots\}$ of finite binary sequences is countable.

Theorem 3.19

The set $\mathbb{N} \times \mathbb{N}$ $(= \mathbb{N}^2)$ of ordered pairs of natural numbers is countable.

Corollary 3.20

The Cartesian product $A \times B$ of two countable sets A and B is countable, i.e., $A \preceq \mathbb{N} \wedge B \preceq \mathbb{N} \implies A \times B \preceq \mathbb{N}$.

Corollary 3.21

The rational numbers $\mathbb{Q}$ are countable.

Theorem 3.22

Let A and A_i for $i \in \mathbb{N}$ be countable sets.

(i) For any $n \in \mathbb{N}$, the set A^n of n-tuples over A is countable.

(ii) The union $\bigcup_{i \in \mathbb{N}} A_i$ of a countable list $A_0, A_1, A_2, \ldots$ of countable sets is countable.

(iii) The set A^* of finite sequences of elements from A is countable.

Uncountability of $\{0,1\}^\infty$

Definition 3.43

Let $\{0,1\}^\infty$ denote the set of semi-infinite binary sequences or, equivalently, the set of functions $\mathbb{N} \to \{0,1\}$.

Theorem 3.23

The set $\{0,1\}^\infty$ is uncountable.

Cantor's Diagonalization Method

$f(i)$	0	1	2	3	4	5	...
$f(0)$	**0**	0	1	1	0	1	...
$f(1)$	1	**1**	0	0	1	0	...
$f(2)$	0	1	**1**	0	0	1	...
$f(3)$	1	0	0	**1**	1	0	...
$f(4)$	0	1	0	1	**0**	0	...
$f(5)$	1	1	0	0	1	**1**	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
α	**1**	**0**	**0**	**0**	**1**	**0**	...

Construct a new sequence α by **flipping** each diagonal bit: $\alpha_i = 1 - f(i)_i$.

Conclusion: α differs from every $f(i)$ at position i, so α cannot be in the list.

Thus, $\{0,1\}^{\mathbb{N}}$ cannot be enumerated.

Uncomputable functions

Definition 3.44

A function $f : \mathbb{N} \to \{0, 1\}$ is called *computable* if there is a program that, for every $n \in \mathbb{N}$, when given n as input, outputs $f(n)$.

Corollary 3.24

There are uncomputable functions $\mathbb{N} \to \{0, 1\}$.

Exercises

Show that if $A \subseteq B$ and $f : B \to A$ is an injection, then $|A| = |B|$.

Let A be a set and let $id_A : A \to A$ the identity function. Prove that $f : A \to A$ is bijective if and only if there exists a function $g : A \to A$ such that

$$g \circ f = id_A \quad \text{and} \quad f \circ g = id_A$$

Exam HS 22

($\star\,\star$) Consider the set

$$S = \{A \subseteq \mathbb{N} \mid A \text{ or } \mathbb{N} \setminus A \text{ is finite}\}.$$

Prove or disprove that the set S is countable. *(5 Points)*

Ex 6.6.3

3. ($\star\,\star$) for all $\ell \in \mathbb{N}$ with $\ell \geq 1$ the set

$$A_\ell := \left\{ f \in \{0,1\}^\infty \;\middle|\; \sum_{i=0}^{k} f(i) \leq \frac{k}{\ell} + 1 \text{ for all } k \in \mathbb{N} \right\}.$$

Hint: Recall that an infinite bit sequence $f \in \{0,1\}^\infty$ can be defined formally as a function $f : \mathbb{N} \to \{0,1\}$.
Hint: For all $\ell \geq 1$, explicitly write an injection from a known uncountable set into A_ℓ.

Exam HS 23

($\star\star$) Consider the set of real numbers that are not roots of polynomials with rational coefficients:

$$A = \{\alpha \in \mathbb{R} \mid p(\alpha) \neq 0 \text{ for all non-zero polynomials } p(x) \in \mathbb{Q}[x]\}.$$

Prove or disprove that the set A is countable. *Hint*: first prove that $\mathbb{Q}[x]$ is countable. *(11 Points)*

Think of $\mathbb{Q}[x]$ as tuples $(a_1, a_2, \ldots)$ where $a_i \in \mathbb{Q}$

First prove that $\mathbb{Q}_n[x]$ is countable.

Every non-constant polynomial of degree n has at most n roots in $\mathbb{R}$

Exam HS 24

e) $(\star\,\star)$ Let X be a set and let ρ be a countable relation on X. **Prove or disprove** that the transitive closure ρ^* of ρ is countable. *(9 Points)*